

RippleBoards: Modeling the Interaction Structure of SDOH to Reduce Premature Mortality in Low-Income U.S. Census Tracts

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Abstract—Premature mortality among low-income Americans is one of the most preventable health inequities in US. Adults below poverty level experience nearly twice the premature mortality rate of higher-income populations. No existing tool simulates synergistic effects of stacking multiple SDOH. RippleBoards tract-level simulation engine enables users to select locations & view projected mortality with confidence intervals.

1 PART A: RESEARCH QUESTION

1.1 Statement

In individuals living in low-income U.S. census tracts (populations below 150% of the Federal Poverty Level), is premature mortality primarily determined by the cumulative burden of independent risk exposures, or by interacting social and structural conditions that modify underlying risk and disease progression pathways, producing differential and potentially amplified responses to targeted interventions?

1.2 Why This Question Matters

This question has direct implications on how public health systems allocate limited resources. Additive and interaction models of SDOH make different empirical predictions & choosing the wrong model can lead to ineffective intervention strategies.

Under *additive model*, premature mortality reflects the sum of independent contributions from housing cost burden, digital exclusion, lack of insurance, poverty, and related structural risks. Each determinant produces a fixed increment of mortality risk regardless of the presence of others. In this framework, single interventions, such as broadband expansion or housing subsidies, produce predictable, context-independent mortality reductions, and resources can be allocated sequentially to the highest marginal-impact determinant.

Under *interaction model*, however, determinants modify and reinforce one another through shared biological and behavioral pathways. Housing instability activates chronic stress responses via HPA-axis dysregulation and sustained cortisol elevation, accelerating cardiovascular disease and type 2 diabetes progression and increasing sensitivity to downstream barriers such as digital exclusion from care. At the same time, digital exclusion prevents

telehealth access and self-management support that could interrupt disease progression accelerated by housing stress. These determinants therefore amplify one another's effects. As a result, broadband expansion alone produces smaller mortality reductions in tracts with severe housing cost burden because key intervention pathways remain partially blocked by concurrent structural instability.

This distinction remains underrepresented in SDOH decision-support tools, which typically assume additive intervention effects. If structural interactions are clinically meaningful, as the evidence reviewed below suggests, then additive models systematically overestimate the benefits of single-factor interventions in multi-disadvantaged communities. RippleBoards addresses this gap by modeling and visualizing interaction effects directly, translating structural interaction evidence into tract-level policy decision support.

2 PART B: LITERATURE REVIEW

2.1 The Scale and Structure of the Premature Mortality Gap

Premature mortality—defined epidemiologically as death before age 75—follows a steep and persistent socioeconomic gradient driven primarily by structural disadvantage rather than biological difference. Kochanek et al. [1], using National Center for Health Statistics vital data, report age-adjusted premature mortality rates of approximately 924 per 100,000 among adults below the federal poverty line compared with 400–500 per 100,000 among higher-income populations. This near-twofold disparity has persisted across two decades of healthcare reform and reflects unequal exposure to upstream structural conditions that shape whether health-protective behaviors and healthcare access are achievable. Assari and Bazargan [2] further demonstrate through mediation analysis of NHANES longitudinal data that the Black–White premature mortality gap is largely explained by socioeconomic position and health-related behaviors, with race functioning primarily as a proxy for structural exposure rather than a biological determinant. This finding establishes that mortality disparities are structurally produced and therefore modifiable through structural intervention. Complementing this population-level evidence, Macias-Konstantopoulos et al. [3] identify institutional disparities in emergency care outcomes for low-income patients, while Caraballo et al. [4] show that healthcare access inequities have remained durable across two decades of reform. Together, these studies establish premature mortality disparities as systemic, persistent, and intervention-relevant.

2.2 The Social-Ecological Model and the Case for Interaction Effects

The Social-Ecological Model (SEM) [5] provides theoretical foundation for understanding SDOH as interacting rather than additive influences operating across nested structural levels. Health outcomes emerge from relationships among household resources, neighborhood infrastructure, institutional access, and policy environments, meaning individual

health behaviors are shaped by structurally conditioned opportunity rather than isolated personal choice. Example adherence to antihypertensive therapy depends simultaneously on housing affordability, broadband enabled telehealth access, transportation availability & health literacy. When these constraints co-occur as they systematically do in low-income census tracts their combined effects exceed the sum of individual pathway contributions. Bailey et al. [6] extend this interaction framework through their analysis of structural racism as a mutually reinforcing system of institutional conditions whose combined effects magnify health inequities beyond what any single determinant would produce independently. At the behavioral level, the Health Belief Model [7] provides a mechanism linking structural interaction to mortality outcomes by showing that multiple simultaneous barriers suppress treatment adherence multiplicatively rather than independently.

2.3 Housing Instability: Pathways Linking Structural Stress to Mortality

Housing instability contributes to premature mortality through interacting biological, environmental, and behavioral pathways. Taylor et al. [8] identify four reinforcing dimensions—physical housing conditions, affordability, residential stability, and neighborhood opportunity—that jointly shape downstream health outcomes. Cost-burdened households are more likely to experience structural housing deficiencies, forced mobility & residence in low-opportunity neighborhoods, meaning housing-related risks compound rather than operate independently. Biologically, housing insecurity activates chronic stress responses through sustained hypothalamic–pituitary–adrenal (HPA) axis activation, increasing cardiovascular disease progression, metabolic dysregulation, immune impairment, and depression-related mortality pathways. Environmental exposures such as mold, lead & inadequate heating further elevate respiratory & cardiovascular risk. Critically, housing cost burden also produces cost-related medication non-adherence, forcing trade-offs between housing expenses & long-term disease management. Intermittent treatment discontinuation produces greater cardiovascular risk than consistent non-treatment because rebound hypertension acts on already-compromised vascular systems. Moss et al. [9] provide strong epidemiological support for these mechanisms, demonstrating a monotonic relationship between housing cost burden and premature cancer mortality that is significantly moderated by Medicaid expansion status—direct evidence of interaction between housing stability and insurance coverage. National data from the Joint Center for Housing Studies [10] further confirm that housing unaffordability has direct relevance as a mortality determinant in low-income census tracts.

2.4 Digital Exclusion as a Structural Super-Determinant

Digital exclusion operates as what Sieck et al. [11] describe as a *super-determinant* of health, conditioning access simultaneously to telehealth, employment opportunities, hous-

ing assistance systems & social services. Rather than contributing an independent mortality increment, broadband absence suppresses the effectiveness of multiple downstream interventions at once making additive simulation assumptions systematically misleading. SAMHSA [12] report that approximately 38% of households earning under \$20K annually lack fixed broadband subscriptions and 40% rely primarily on constrained mobile data plans, limiting effective telehealth participation. McCall et al. [13] characterize this pattern as digital redlining—structural concentration of inadequate internet infrastructure in communities already affected by housing segregation—explaining why broadband deficits & housing instability frequently co-occur in disadvantaged census tracts. Okoye et al. [14] further demonstrate that neighborhood broadband availability predicts telehealth utilization even after controlling for individual socioeconomic characteristics, confirming a contextual infrastructure effect. Technology Acceptance Model(TAM) [15] explains this mechanism behaviorally: unreliable infrastructure prevents perceived ease of use, producing persistent disengagement from digital care pathways even after access improves.

2.5 Telehealth Inequities and Infrastructure Dependence

The COVID-19 pandemic created a large-scale natural experiment demonstrating the infrastructure dependence of digital health interventions. Pandit et al. [16] show that Medicare telehealth utilization increased most in counties with stronger broadband capacity despite nationwide coverage expansion, confirming infrastructure—not program availability—as the binding constraint. Similarly, Nguyen et al. [17] identify persistent telehealth disparities among low-income racial and ethnic minority populations driven by device access, digital literacy, and trust barriers that reinforce one another. D’Amico et al. [18] show that telehealth gains for low-income patients were smaller and less durable than those observed among higher-income populations, while Clare [19] demonstrates that the pandemic made the health consequences of the digital divide directly measurable. Eubanks [20] further cautions that technology-enabled public systems risk reproducing inequities when deployed without structural safeguards.

2.6 Synthesis: Themes, Divergences, and the Research Gap

Three cross-cutting themes emerge consistently across the literature. First, premature mortality disparities in low-income communities are structurally produced rather than behaviorally determined [1, 2, 6, 8, 11], yet their quantitative interaction structure remains largely unmodeled for decision-making contexts. Second, digital health interventions without infrastructure prerequisites reproduce rather than reduce disparities [16–19], implying that intervention effectiveness depends on concurrent structural conditions. Third, existing SDOH datasets remain inaccessible to non-technical policymakers despite containing actionable information [12, 13], creating a persistent information-to-action gap.

Although the literature differs on whether housing stability or broadband access is the primary intervention lever [8, 9, 11], our expert interviews resolve this divergence through the interaction model: neither determinant is primary in isolation; their interaction determines mortality outcomes, and relative priority varies by census-tract conditions.

Research Gap. Existing research establishes that premature mortality in low-income census tracts is produced by interacting structural conditions rather than additive independent risks. However, no prior work operationalizes these interaction effects into a tract-level multi-intervention simulation tool accessible to policymakers. RippleBoards addresses this translation gap by making the interaction premium of bundled interventions visible for evidence-based decision support.

3 PART C: KEY INFORMANT INTERVIEW FINDINGS

3.1 Methodology and Informant Selection

To validate the interaction hypothesis and RippleBoards’ design from practitioner perspectives, we conducted two structured expert interviews in March 2026 using a Google Forms questionnaire. Informants were selected to triangulate the research question from complementary vantage points: one representing upstream pharmaceutical access and health equity policy, and one representing downstream population-health analytics and implementation constraints. Informed consent was obtained from both participants. Full interview questions are provided in Appendix C.

Informant 1: Abha Dhamiwal, Life Sciences and Pharma Researcher at Ernst & Young(8 years experience), specializes in financial and structural barriers to medication adherence in low-income populations.

Informant 2: Shubham Khode, Senior Data Scientist at CVS Health(6 years experience), works on population-health analytics & predictive modeling of healthcare utilization including digital-health program performance across heterogeneous census-tract contexts.

3.2 Findings and Literature Integration

3.2.1 Confirmation of Structural Interaction: Cost-Related Non-Adherence

Dhamiwal rated the influence of social determinants on patient outcomes as “5 out of 5—extremely significant” and described recurring medication-rationing behaviors among patients facing severe housing cost burden, including dose splitting, delayed refills, and treatment discontinuation during high-expense months. These behaviors represent rational responses to constrained household budgets rather than failures of motivation, directly linking housing instability to downstream disease progression.

Her account aligns closely with the pathophysiological mechanism described in the litera-

ture review: discontinuation after partial treatment produces rebound hypertension acting on already-compromised vascular systems, accelerating cardiovascular risk beyond that of consistently untreated patients. This observation confirms the SEM prediction [5] that exosystem constraints such as housing affordability cascade into microsystem health behaviors through financial capacity limits. Importantly, Dhamiwal emphasized that this effect intensifies when patients also lack broadband access for telehealth follow-up, identifying a practitioner-observed interaction between housing stress and digital exclusion that amplifies mortality risk beyond additive expectation.

3.2.2 Elaboration of Synergistic Intervention Effects

When asked whether medication subsidies alone substantially reduce premature mortality in high-burden census tracts, Dhamiwal emphasized the greater effectiveness of combined interventions: *“A drug subsidy is a great start, but it is much more effective if you stack it. If RippleBoards shows that a drug subsidy plus a transportation grant leads to a higher mortality reduction than the subsidy alone, you give policymakers a roadmap for a successful intervention.”*

This response clarifies the decision-support function of interaction modeling: policymakers require quantitative estimates of the *interaction premium*, the additional mortality reduction achieved by combining interventions beyond their independent effects. This insight refines the prototype requirement from supporting intervention stacking to explicitly estimating and communicating the marginal benefit of bundled strategies, a capability absent from existing SDOH dashboards [6, 8, 11].

3.2.3 Confirmation of Digital Infrastructure as a Structural Prerequisite

Khode confirmed the super-determinant role of broadband infrastructure through organizational evidence from CVS Health, where digital-health programs consistently underperform in census tracts lacking reliable internet access. He characterized broadband as the “front door of modern medicine” reinforcing the contextual infrastructure effect identified by Okoye et al. [14] and predicted by the Technology Acceptance Model [15]. Programs with identical clinical design produced substantially weaker outcomes in low-broadband tracts demonstrating that infrastructure absence constrains intervention effectiveness independent of clinical intent.

Khode also recommended displaying 95% confidence intervals alongside projected mortality changes: *“Since you are using static census data to create a dynamic simulation, users must understand the uncertainty in prediction.”* This recommendation directly informed Requirement 5 by ensuring simulation outputs communicate uncertainty transparently & discourage categorical allocation decisions based solely on point estimates.

3.2.4 Resolution of the Primacy Debate and Validation of the Interaction Model

Both informants independently resolved the housing-versus-broadband primacy debate reflected in the literature by emphasizing that neither determinant operates as a dominant driver in isolation. Khode noted that transportation and broadband function as enabling infrastructure for clinical interventions, while Dhamiwal observed that medication subsidies produce their greatest mortality benefit only when housing stability and digital access allow sustained adherence behaviors.

Together, these perspectives confirm the interaction premise of our research question: intervention effectiveness depends on the structural context in which it is deployed. In high housing-stress & low-broadband tracts, the mortality benefit of single-factor interventions is attenuated by concurrent barriers, whereas combined structural improvements produce larger-than-additive effects. This convergence directly supports the need for a census-tract-level multi-intervention simulation engine that makes interaction effects visible to policymakers and distinguishes between additive & synergistic intervention strategies.

4 PART D: DESIGN RATIONALE AND PROTOTYPE

4.1 Design Requirements

These 5 core design requirements(DR) emerged from the literature and expert interviews

DR-1: Census-Tract-Level Geographic Granularity. The literature shows that health disparities operate at hyperlocal spatial scales rather than county-level aggregates [3, 16]. Adjacent census tracts can differ by more than 200 deaths per 100,000 in age-adjusted premature mortality, reflecting clustered structural disadvantage that county summaries obscure. Khode confirmed this pattern from CVS Health implementation experience. RippleBoards therefore operates at CDC PLACES census-tract resolution, enabling tract-level visualization of SDOH exposure and projected intervention impact where structural policies are implemented.

DR-2: Multi-Intervention Stacking with Interaction-Aware Modeling. Both the literature and expert interviews show that SDOH interventions interact rather than operate independently [6, 9, 11]. Additive models overestimate single-factor returns and underestimate bundled intervention benefits in structurally disadvantaged tracts. RippleBoards therefore implements an interaction-aware stacking engine that models cross-term effects between interventions rather than treating determinants as isolated contributors.

DR-3: Before/After Trajectory Visualization with Ripple-Structured Output. Khode identified time-series Before/After mortality trajectories as the format most likely to influence funding decisions. Structural interventions such as housing stabilization and

broadband expansion produce benefits that accumulate over time, which single summary statistics cannot capture. Health communication research similarly shows that visual projections improve interpretability for non-technical decision-makers [7]. RippleBoards therefore presents intervention impact through trajectory-based visualization supported by ripple-structured causal outputs.

DR-4: Accessibility for Non-Data-Scientists. SAMHSA and McCall et al. identify inaccessible data infrastructure as a primary barrier to evidence-based structural intervention [12, 13]. Both informants confirmed that intended users—including city health administrators, county public health officers, and grant writers—typically lack statistical programming expertise. RippleBoards therefore translates multi-SDOH interaction modeling into an interface requiring no technical training.

DR-5: Uncertainty Quantification Through Confidence Intervals. Simulation based on observational data necessarily involves uncertainty. Khode emphasized that presenting point estimates alone risks overconfident allocation decisions. RippleBoards therefore displays 95% confidence intervals alongside projected mortality trajectories, allowing users to assess whether alternative intervention scenarios are meaningfully distinguishable.

Design Alternatives Considered and Rejected

Alternative A: Static SDOH Dashboard. Tools such as CDC SVI Explorer and County Health Rankings provide descriptive visualization of structural risk factors but do not simulate intervention effects. While they identify disparities they cannot estimate how outcomes change under tract-level or combined intervention scenarios.

Alternative B: Machine Learning Mortality Prediction API. A machine-learning prediction API would offer strong statistical performance but fails three requirements. It violates DR-4 by requiring programming expertise, lacks the transparency needed for policy justification, and risks reproducing structural inequities when deployed without interpretability safeguards [20].

4.2 RippleBoards: Prototype Features and Innovations

RippleBoards is the only design alternative that satisfies all five design requirements simultaneously while directly addressing premature mortality disparities in low-income census tracts through interaction-aware structural intervention modeling. Its primary innovation—computing and visualizing the *interaction premium* of bundled interventions over single-factor strategies using the interaction model in Equation 1—operationalizes the central finding of both the literature review and expert interviews: that structural determinants act synergistically rather than independently. Supporting storyboard iterations

illustrating this design evolution are provided in the appendix.

RippleBoards is deployed at <https://rippleboard-sdoh.web.app/> and accessible from any browser without installation, supporting practical use by non-technical public health administrators (DR-4). The system integrates CDC PLACES tract-level health indicators (ACS 2017–2021) with ACS 5-year socioeconomic estimates via CENSUS-TRACT_GEOID, enabling tract-resolution simulation of structural intervention effects on premature mortality. Figure 2 shows the interface; Table 1 in the appendix, summarizes key dataset columns and their roles in the simulation engine.

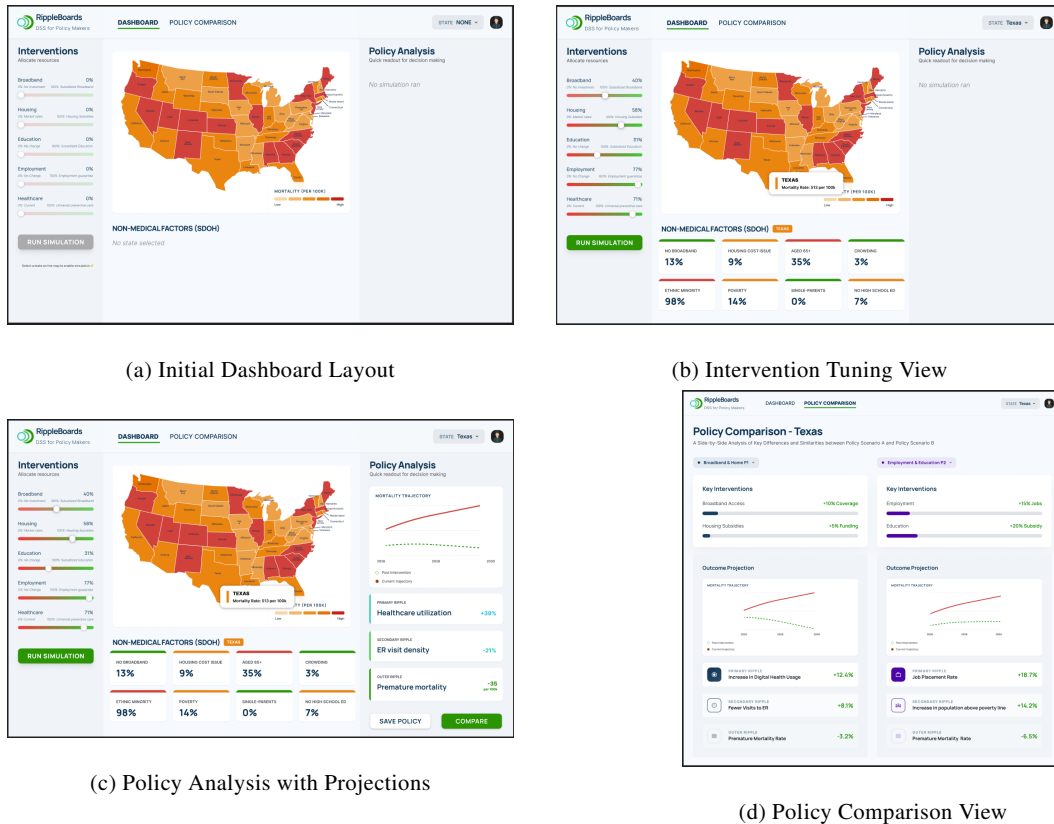


Figure 1—Figma design alternatives explored prior to HTML implementation. These iterations informed layout decisions, interaction flow & visualization choices in the final system.

Layer 1: Geographic Selection and SDOH Profile Loading (Implements DR-1). Center panel presents an interactive US map at census-tract resolution, color-coded by current premature mortality rate (green <350 to red ≥ 550 per 100K). Selecting tract reveals structural SDOH profile including housing burden, unemployment, broadband deficit, poverty rate, educational attainment, aging population share, crowding index & minority popula-

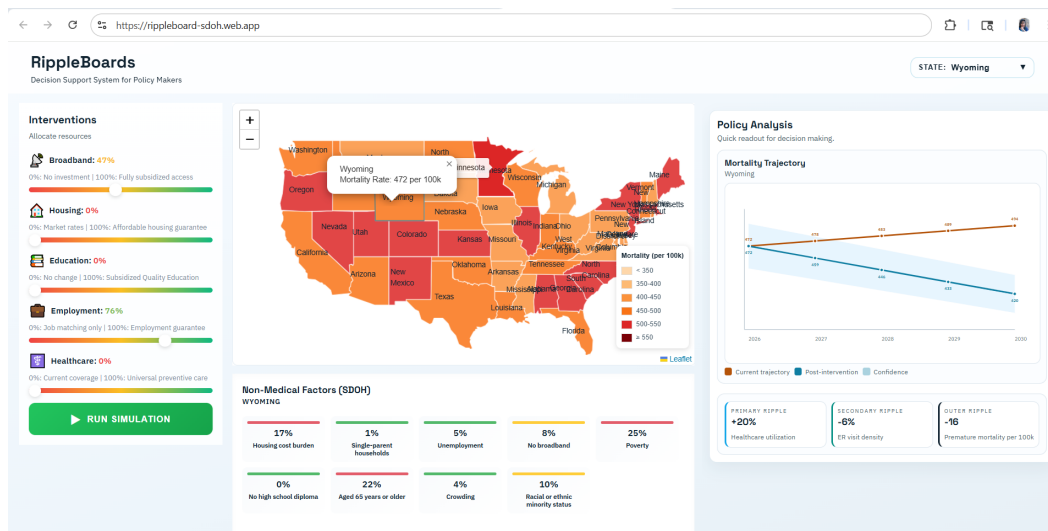


Figure 2—RippleBoards **Left:** multi-intervention stacking controls
Center: census-tract mortality map with tract-level SDOH profile.
Right: policy analysis showing mortality trajectory & ripple outputs

tion share. This tract-level diagnostic view allows users to identify locally concentrated structural risks & prioritize interventions where mortality reduction potential is greatest.

Layer 2: Multi-Intervention Stacking Engine (Implements DR-2). The left panel provides five intervention sliders representing broadband expansion, housing stabilization, education access improvement, employment support, and preventive healthcare coverage (each scaled 0–100%). Projected mortality outcomes are computed using an interaction-aware simulation model:

$$\Delta M = \sum_j \beta_j \cdot \delta_j + \sum_{j < k} \gamma_{jk} \cdot \delta_j \cdot \delta_k \quad (1)$$

where β_j represents main-effect intervention coefficients derived from literature-supported pathways, δ_j represents intervention intensity, and γ_{jk} represents interaction coefficients capturing synergistic structural effects.

Layer 3: Ripple-Structured Impact Reporting (Implements DR-3 and DR-5). The right panel presents simulation outputs through a three-tier ripple structure linking structural interventions to downstream mortality outcomes. The *Primary Ripple* reports immediate structural changes (e.g., “+20% healthcare utilization”), the *Secondary Ripple* reports intermediate clinical effects (e.g., “−6% ER visit density”), and the *Outer Ripple* reports projected premature mortality change (e.g., “−16 deaths per 100,000”). Above these indicators, a five-year mortality trajectory (2026–2030) compares baseline and inter-

vention scenarios with shaded 95% confidence bands, enabling policymakers to interpret both projected impact magnitude and uncertainty.

4.3 Potential to Positively Address the Health Disparity

RippleBoards addresses premature mortality disparities through following 3 mechanisms

First, it closes the *information-to-action gap* identified by McCall et al. and SAMHSA [12, 13]. While epidemiological evidence documenting SDOH interaction effects already exists, policymakers currently lack tract-level tools translating that evidence into intervention-relevant projections. RippleBoards operationalizes this evidence into an accessible, interaction-aware simulation interface.

Second, it makes the *synergistic benefit of bundled interventions visible and quantifiable*. As confirmed in practitioner interviews, structural interventions rarely operate independently in practice. RippleBoards enables direct comparison between single-factor and multi-factor intervention scenarios within the same tract, allowing policymakers to evaluate whether combined strategies produce statistically meaningful mortality improvements.

Third, it enables *evidence-based geographic targeting* at the resolution where structural interventions are implemented. By operating at census-tract scale and incorporating tract-specific SDOH exposure profiles, RippleBoards supports identification of locations where overlapping broadband deficits and housing burden produce the largest projected mortality gains under coordinated intervention strategies. This targeting improves allocation efficiency while prioritizing structurally disadvantaged communities.

4.4 Ethical Considerations for Real-World Deployment

- **Algorithmic bias and model validity-** RippleBoards relies on observational CDC PLACES data that may undercount disease burden in underserved communities due to diagnostic access gaps and survey nonresponse bias. Responsible deployment requires transparent assumptions consistent with recommendations for ongoing disparate-impact auditing of public-sector algorithms [20].
- **Risk of misuse in resource allocation-** Projected mortality reductions across census tracts could be misinterpreted as optimization signals rather than decision-support inputs, potentially disadvantaging communities with complex structural barriers.
- **Data ethics and re-identification risk-** Although RippleBoards uses aggregate census-tract indicators, combining multiple structural variables can increase indirect re-identification risk when linked with external datasets.
- **Community agency and participatory design-** Effective deployment should incorporate participatory co-design with affected communities to ensure intervention priorities reflect lived structural conditions rather than administrative data alone.

4.4.1 Team Member Contributions

Chethana Prasad Kabgere: Developed the initial proof-of-concept for the research problem and led the overall design of RippleBoards, contributing significantly to both the system concept and implementation direction. Was actively involved in key deliverables, including presenting the project at Mediaspaces, contributing to the research paper writing, and designing the prototype. My contributions were sustained throughout the semester and had a direct impact on shaping both the technical and conceptual outcomes of the project.

Breejesh Rathod: Played a crucial role in executing the core idea of the project and translating it into actionable components. He contributed significantly to research, literature survey and also took the key informant interview that helped us in refining our research direction. He led the efforts for initial Figma design of Rippleboards. His involvement helped strengthen both the conceptual grounding and practical progression of the project, and he was consistently engaged throughout.

Shivam Chhabra: made significant contributions to the literature research and documentation aspects of the project as well as proofreading final submission. Have played a key role in developing the storyboard and preparing the project presentation, ensuring that ideas are clearly and effectively conveyed. An active team member throughout, contributing to discussions and helping maintain steady progress.

Neel Sharma: Strong contributions to both the literature survey and the Figma prototype design. He helped translate conceptual ideas into tangible interface designs, which was critical for visualizing the system. His work on the prototype added clarity to the project's usability and interaction flow, and he remained actively involved in supporting the team's progress.

All four members collaborated on research question finalization, project conceptual framing, and prototype design decisions throughout the semester.

4.4.2 Prototype Links

Live Web Prototype: <https://rippleboard-sdoh.web.app/>

Figma Prototype: <https://www.figma.com/proto/COKPyYUObsKAblBsN74PhG/Rippleboard?node-id=0-1&t=m9003chPlGUnPqds-1>

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APPENDIX

A DATASET

Table 1—Key CDC PLACES and ACS Dataset Columns in Ripple-Boards

Column / Measure	Source / Type	Role in Simulation
AAPMORT	CDC PLACES / Outcome	Primary dependent variable: age-adjusted premature mortality rate per 100,000
BPHIGH_CrudePrev	CDC PLACES / Health	High BP prevalence; intermediate marker on housing–mortality pathway
ACCESS2_CrudePrev	CDC PLACES / SDOH	Uninsured adults; healthcare access inequality indicator
BROADBND	ACS / SDOH-Digital	Broadband subscription rate; primary digital access determinant
HOUSING_COST_BURDEN	ACS / SDOH-Housing	Housing affordability exposure variable
POVERTY_RATE	ACS / Socioeconomic	Root structural risk moderator of intervention effects
CENSUSTRACT_GEOID	ACS + PLACES	Spatial join key at census-tract resolution

B STORYBOARD

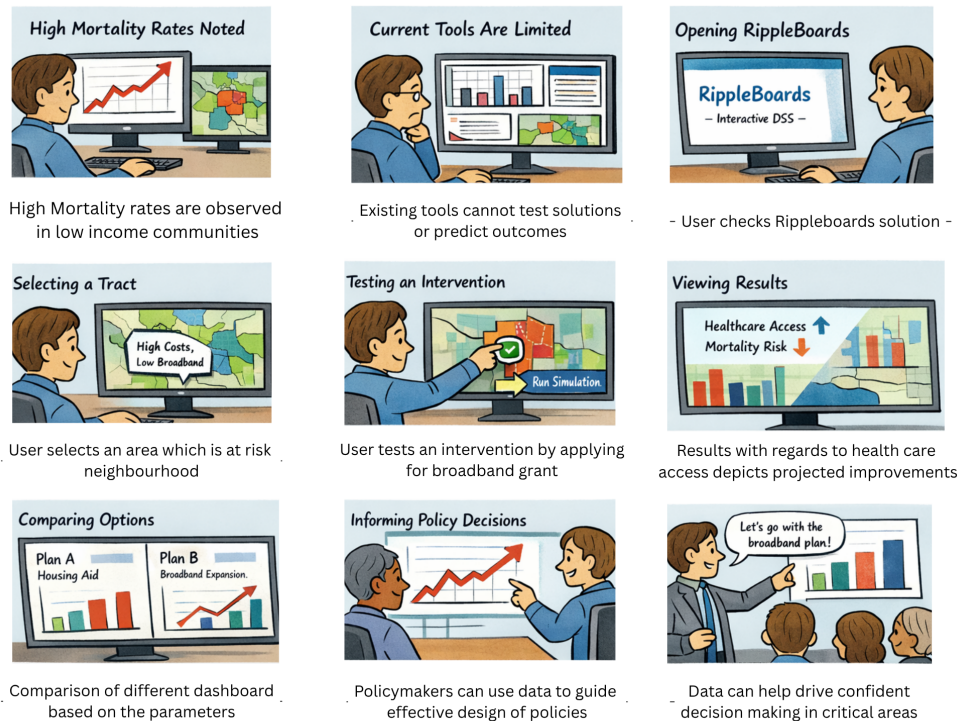


Figure 3—Rippleboards Storyboard

C KEY INFORMANT INTERVIEW QUESTIONS

Questions administered via Google Forms to both informants (March 2026). Consent to participate and use responses for academic research was obtained.

Section 1: Professional Background

- Q1. Describe your current role, organizational context, and tenure in this field.
- Q2. How does your work intersect with healthcare access, housing stability, or digital equity for low-income populations?

Section 2: Social Determinants and Health Outcomes

- Q3. On a 1–5 scale, how significantly do social determinants influence the health outcomes of patients or populations you work with?
- Q4. What is the single most impactful structural barrier to equitable health outcomes for low-income populations?

- Q5. Have you observed housing instability directly affecting medication adherence? Describe the mechanism.
- Q6. Do structural barriers tend to act independently (additive risk) or interactively (each amplifying the others)? Provide an example.

Section 3: Digital Access and Telehealth

- Q7. Have broadband access or digital literacy deficits prevented patients from using telehealth effectively in your experience?
- Q8. Has telehealth expansion reduced or reproduced healthcare disparities for low-income populations?
- Q9. What digital infrastructure investment would most improve health outcomes in the low-income communities you analyze?

Section 4: Decision-Support and Policy Simulation

- Q10. Which SDOH intervention would you model first in RippleBoards, and why?
- Q11. How difficult is it to predict a proposed SDOH intervention's mortality impact before implementation?
- Q12. What visualization format would most improve the budget arguments or grant applications you prepare?
- Q13. Would a Before/After mortality trajectory change how your organization prioritizes resource allocation?

Section 5: Critical Feedback on RippleBoards

- Q14. Would RippleBoards be useful for policymakers or practitioners in your field? Why or why not?
- Q15. What would you change or add for your professional context?
- Q16. What risks of simulation-based SDOH policy planning should the tool address?
- Q17. Which intervention combination would produce the greatest mortality reduction in a low-broadband, high-housing-burden census tract? Would the combination exceed the sum of individual effects?

GENERATIVE AI STATEMENT

We have used generative AI to prepare this assignment. We used ChatGPT for image generation of Storyboard. The prompt used was “generate storyboard images on using rippleboards as decision-support system to address premature mortality in low-income communities”